

MAT 21D Discussion 3 Notes

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§I Warm-up

The iterated integral

$$\int_0^{2\pi} \int_0^3 r^2 dr d\theta$$

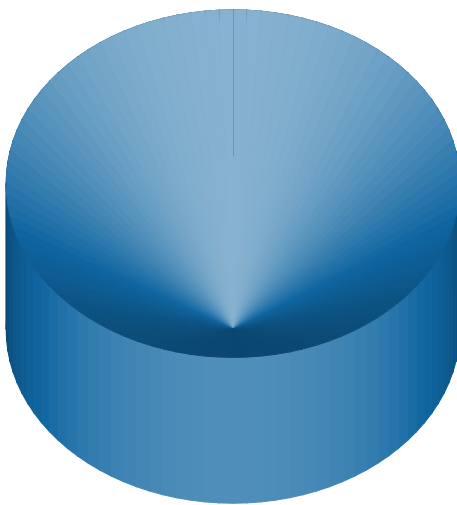
describes the volume of a solid. Describe the solid to your group in words or with a sketch.

Solution. Note that because $dA = r dr d\theta$, we have

$$\int_0^{2\pi} \int_0^3 r^2 dr d\theta = \iint_R r dA$$

where R is the disk of radius 3 centered at the origin in the xy -plane. Thus, the solid is the region above the disk and below the graph of $z = r = \sqrt{x^2 + y^2}$.

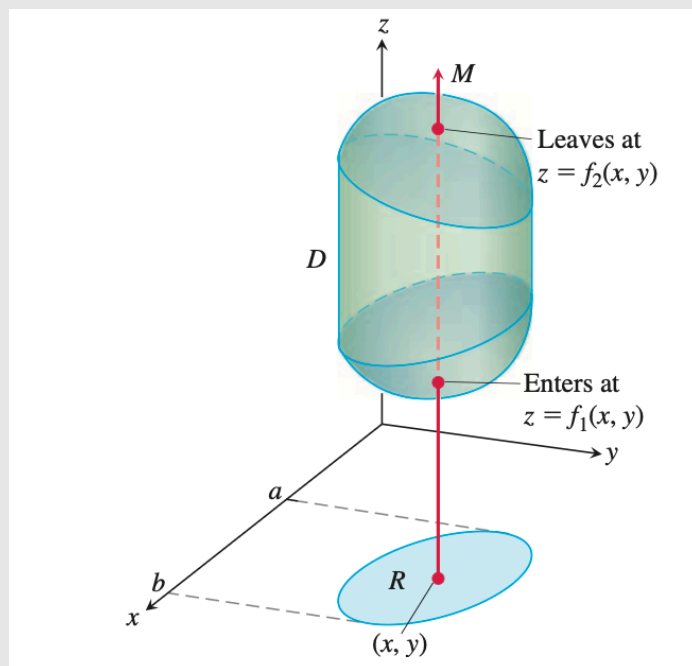
We can describe it as a “crown shaped” region obtained by removing a cone from the cylinder of radius 3 and height 3.



§II Triple integrals

If a region D has shadow R in the xy -plane and is described by $f_1(x, y) \leq z \leq f_2(x, y)$, then the triple integral over D can be computed as

$$\iiint_D F(x, y, z) dV = \iint_R \int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz dA.$$



Furthermore, if R is described by $a \leq x \leq b$ and $g_1(x) \leq y \leq g_2(x)$, then

$$\iiint_D F(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{f_1(x, y)}^{f_2(x, y)} F(x, y, z) dz dy dx.$$

The first part of the above theorem is the important part. The second part is just an application of Fubini's theorem to the double integral. Fubini's theorem is not the only way to compute double integrals. You also know about polar coordinates, and if we switch out the second part of the theorem with evaluating by polar coordinates, we will get a formula for triple integrals in cylindrical coordinates.

Let D denote the solid region lying above the square with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 2, 0)$, and $(1, 2, 0)$ in the xy -plane and bounded above by the graph of the function $z = f(x, y) = x^2$.

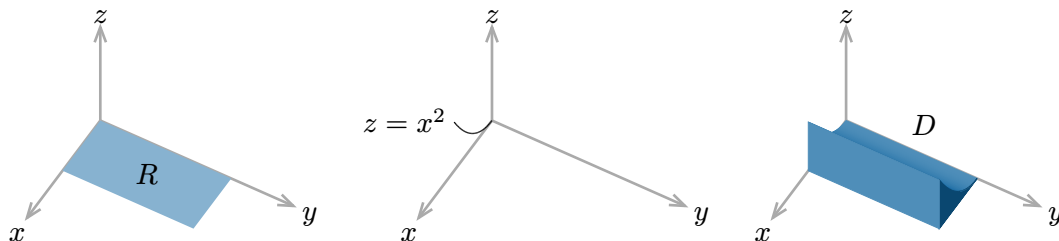
1. Compute the volume of the region D by evaluating the triple integral

$$\iiint_D 1 dV.$$

2. Evaluate the triple integral

$$\iiint_D x dV.$$

Solution. The shadow of the region D in the xy -plane is the rectangle given by $0 \leq x \leq 1$ and $0 \leq y \leq 2$, and the bounds of z are given by $0 \leq z \leq x^2$.



1. The volume of the region D is given by

$$\int_0^1 \int_0^2 \int_0^{x^2} 1 \, dz \, dy \, dx = \int_0^1 \int_0^2 x^2 \, dy \, dx = \int_0^1 2x^2 \, dx = \frac{2}{3}.$$

2. The value of the triple integral is

$$\begin{aligned} \int_0^1 \int_0^2 \int_0^{x^2} x \, dz \, dy \, dx &= \int_0^1 \int_0^2 x^3 \, dy \, dx \\ &= \int_0^1 2x^3 \, dx \\ &= \frac{1}{2}. \end{aligned}$$

Sometimes you need to be flexible with variations of the theorem for triple integrals. You may want to cast the shadow to a different plane like the yz -plane or the xz -plane. For example, here is variation of the theorem about casting the shadow to the yz -plane.

If a region D has shadow R in the yz -plane and is described by $f_1(y, z) \leq x \leq f_2(y, z)$, then the triple integral over D can be computed as

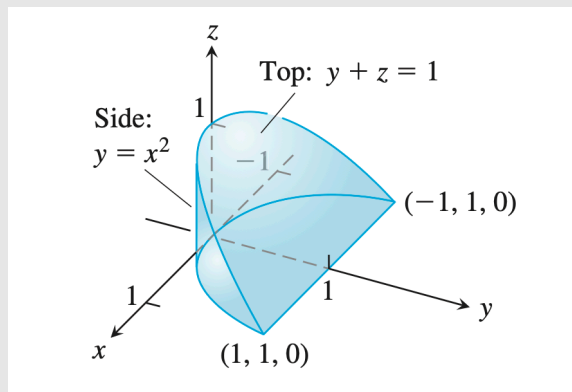
$$\iiint_D F(x, y, z) \, dV = \iint_R \int_{f_1(y, z)}^{f_2(y, z)} F(x, y, z) \, dx \, dA.$$

When evaluating the double integral over R , be mindful that R is in the yz -plane, so dA is $dy \, dz$ or $dz \, dy$.

The triple integral can be thought of as a bridge between all the iterated integrals in different orders.

(Exercise 15.5.21) Here is the region of integration of the integral

$$\int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} dz \, dy \, dx.$$



Rewrite the integral as an equivalent iterated integral in the order

1. $dz dx dy$
2. $dx dy dz$
3. $dx dz dy$
4. $dy dz dx$
5. $dy dx dz$.

Choose two of the iterated integrals and evaluate them to verify that they give the same value.

Solution.

1. The shadow of the region in the xy -plane is given by $-1 \leq x \leq 1$ and $x^2 \leq y \leq 1$. Equivalently, it is given by $0 \leq y \leq 1$ and $-\sqrt{y} \leq x \leq \sqrt{y}$. Thus, the integral in the order $dz dx dy$ is

$$\begin{aligned}
 \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} \int_0^{1-y} dz dx dy &= \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} (1-y) dx dy \\
 &= \int_0^1 2\sqrt{y}(1-y) dy \\
 &= 2 \int_0^1 (y^{\frac{1}{2}} - y^{\frac{3}{2}}) dy \\
 &= 2 \left(\frac{2}{3} - \frac{2}{5} \right) \\
 &= \frac{8}{15}.
 \end{aligned}$$

2. The bounds of x described as functions of y and z are $-\sqrt{y} \leq x \leq \sqrt{y}$. Thus, the triple integral is equal to

$$\iint_R \int_{-\sqrt{y}}^{\sqrt{y}} dx dA$$

where R is the shadow of the region in the yz -plane.

This shadow R is the triangle bounded by the y and z axes and $y + z = 1$. We can describe R as $0 \leq z \leq 1$ and $0 \leq y \leq 1 - z$. Thus, the integral in the order $dx dy dz$ is

$$\begin{aligned}
\int_0^1 \int_0^{1-z} \int_{-\sqrt{y}}^{\sqrt{y}} dx dy dz &= \int_0^1 \int_0^{1-z} 2\sqrt{y} dy dz \\
&= \int_0^1 \left(\frac{4}{3}\right) (1-z)^{\frac{3}{2}} dz \\
&= -\int_1^0 \left(\frac{4}{3}\right) u^{\frac{3}{2}} du \\
&= -\left[\left(\frac{4}{3}\right) \left(\frac{2}{5}\right) u^{\frac{5}{2}}\right]_1^0 \\
&= \frac{8}{15}.
\end{aligned}$$

We have used the substitution $u = 1 - z$ and thus $du = -dz$.

3. We also start with

$$\iint_R \int_{-\sqrt{y}}^{\sqrt{y}} dx dA.$$

Now, we describe R as $0 \leq y \leq 1$ and $0 \leq z \leq 1 - y$. Thus, the integral in the order $dx dz dy$ is

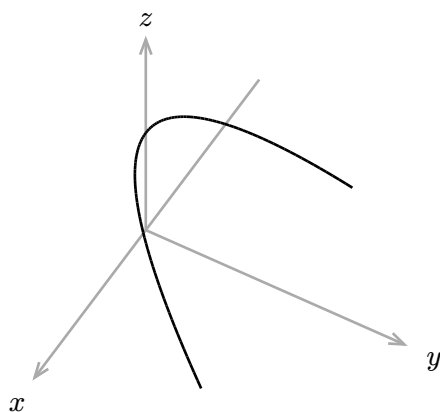
$$\begin{aligned}
\int_0^1 \int_0^{1-y} \int_{-\sqrt{y}}^{\sqrt{y}} dx dz dy &= \int_0^1 \int_0^{1-y} 2\sqrt{y} dz dy \\
&= \int_0^1 2\sqrt{y}(1-y) dy \\
&= 2 \int_0^1 \left(y^{\frac{1}{2}} - y^{\frac{3}{2}}\right) dy \\
&= \frac{8}{15}.
\end{aligned}$$

We have already done the last integral in part 1.

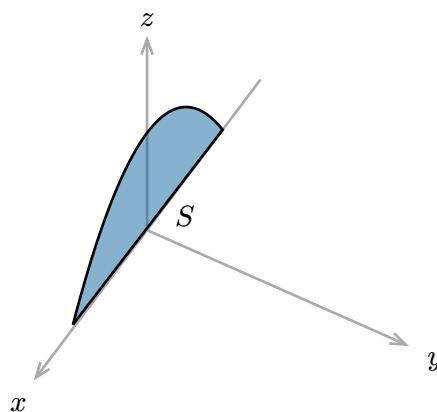
4. The bounds of y described as functions of x and z are $x^2 \leq y \leq 1 - z$. Thus, the triple integral is equal to

$$\iint_S \int_{x^2}^{1-z} dy dA$$

where S is the shadow of the region in the xz -plane. We can find the bounds of S by projecting the intersection of $y = x^2$ and $y + z = 1$ to the xz -plane.



(a) Intersection of $y = x^2$ and $y + z = 1$.



(b) Shadow S in the xz -plane. The boundary is obtained by projecting the intersection in subfigure (a) onto the xz -plane.

This can be done by finding the relationship between x and z when we have both $y = x^2$ and $y + z = 1$. Doing so gives us $x^2 + z = 1$. Thus, we can describe S as $-1 \leq x \leq 1$ and $0 \leq z \leq 1 - x^2$. Thus, the integral in the order $dydzdx$ is

$$\begin{aligned} \int_{-1}^1 \int_0^{1-x^2} \int_{x^2}^{1-z} dy \, dz \, dx &= \int_{-1}^1 \int_0^{1-x^2} (1 - z - x^2) \, dz \, dx \\ &= \int_{-1}^1 \left((1 - x^2)^2 - \frac{1}{2}(1 - x^2)^2 \right) dx \\ &= \frac{1}{2} \int_{-1}^1 (1 - x^2)^2 dx \\ &= \frac{1}{2} \int_{-1}^1 (1 - 2x^2 + x^4) dx \\ &= \frac{1}{2} \left[x - \left(\frac{2}{3}\right)x^3 + \left(\frac{1}{5}\right)x^5 \right]_{-1}^1 \\ &= 1 - \frac{2}{3} + \frac{1}{5} \\ &= \frac{8}{15}. \end{aligned}$$

5. Finally, we can also describe S as $0 \leq z \leq 1$ and $-\sqrt{1-z} \leq x \leq \sqrt{1-z}$. Thus, the integral in the order $dydx dz$ is

$$\begin{aligned} \int_0^1 \int_{-\sqrt{1-z}}^{\sqrt{1-z}} \int_{x^2}^{1-z} dy \, dx \, dz &= \int_0^1 \int_{-\sqrt{1-z}}^{\sqrt{1-z}} (1 - z - x^2) \, dx \, dz \\ &= \int_0^1 \left[(1 - z)x - \frac{1}{3}x^3 \right]_{-\sqrt{1-z}}^{\sqrt{1-z}} dz \\ &= \int_0^1 \left(\frac{4}{3} \right) (1 - z)^{\frac{3}{2}} dz \\ &= \frac{8}{15}. \end{aligned}$$

We have done the last integral in part 2.